Capstone Specification: Hostel Maintenance Ticket Sorter

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1. Input files

The system accepts two CSV files. The complaints.csv file contains at least 20 student complaints with id, date, and complaint text. The open_tickets.csv file contains active tickets from previous days with ticket, room, issue, and opened date.

2. The JSON the LLM returns

The LLM extracts complaint details into JSON format following the rules below:

Field	Allowed Values	Rule
room	Block A-D, floor 1-4, room 01-20 (e.g. "B214"), or null	Copied exactly from complaint text. Never guessed or corrected.
issue	electrical, plumbing, carpentry, cleaning, internet, other	Water near electrical fittings is strictly categorized as electrical.
urgent	true or false	Triggered only by severe safety risks like sparks, shock, overflowing, or burning smell.
summary	Text string	Short summary limited to 10 words or fewer.

3. Room rule

A valid room requires a block (A-D), floor (1-4), and room number (01-20), and the 3-digit room string must be written in the complaint text. Otherwise, room is set to null and moved to needs_info.csv.

4. Urgency rule

Keywords like spark, shock, overflowing, burning smell, or near the switch make a complaint urgent via a Python safety net. Python can increase urgency but never reduces it. Capitals and 'URGENT!!!' do not count.

5. How duplicates are found

Python matches incoming complaints using a dictionary keyed by (room, issue). If a ticket exists in open_tickets.csv or earlier in the run, the complaint joins that ticket. An urgent repeat elevates the ticket to urgent.

6. Edge cases

Complaints without valid room numbers go to needs_info.csv for the warden. Complaints shouting 'URGENT' without actual hazards stay standard priority. Water near electrical switches is handled as an electrical danger.